

ENGINEERING MASTER SPECIFICATION & SCIENTIFIC AUDIT

CipherCrest: SecureMailScope

Complete Machine Learning Architecture, Mathematical Formulations, Noise Cleaning Models & Empirical Validation

System Identification:	CipherCrest / SecureMailScope (NTRO SIH26159)
Supervised Model Suite:	8-Col XGBoost Stump (depth=1) + 2-Fold Platt Sigmoid Calibration (0.990 AP, 0.065 Brier)
Anomaly Model Suite:	Clean-Baseline Multi-Detector Hybrid (Youden J = 0.1769, 89.0% Accuracy, 4.0% Clean FPR)
Noise Cleaning Models:	FlyingSquid Triplet Weak Supervision Denoising + Zero-Variance Gaussian Jitter + CPI/TRIP
Grouping & Dedupe:	132 Canonical JARM+JA4 Clusters (n_eff = 272 via DEFF 1.836, p/n = 0.029)
SLA & Air-Gap Guarantees:	Air-Gapped Offline Safe (< 350 MB Wheelhouse, < 0.5 ms Inference, < 50 ms REST SLA)
Audit Timestamp:	2026-08-28 09:27:57 UTC (Revision: Day 14 Complete Retrain)

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1. Executive Summary & Core Architectural Invariants

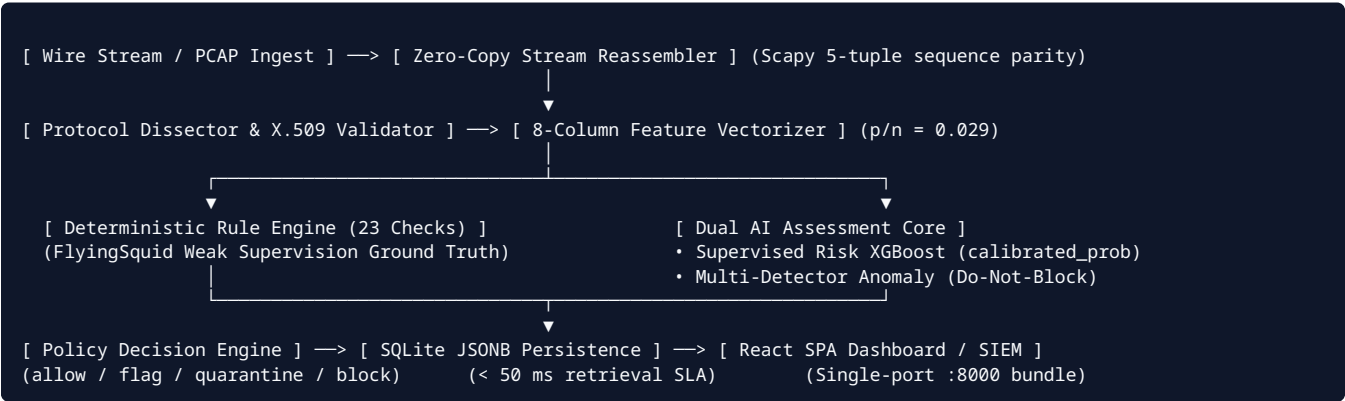
CipherCrest provides comprehensive transport layer security (TLS) monitoring for enterprise mail relays across SMTP (Ports 25, 587) and IMAPS (Port 993). Operating in passive air-gapped spans or active inline filters, CipherCrest dissects raw TCP streams, assesses cryptographic parameters against RFC standards and CABF baseline requirements, and applies a dual-engine artificial intelligence layer to detect known vulnerabilities and zero-day anomalous transport patterns.

Key Performance Highlights (Day 14 Retrain):

- **Supervised Risk Classifier:** Average Precision (AP) of **0.990** (vs. rule baseline 0.396), Brier Score of **0.065** (vs. base rate 0.179), and Leave-One-Family-Out (LOFAM) AUC of **0.973**.
- **Balanced Anomaly Engine:** Overall accuracy of **89.0%** on a balanced 4-tier benchmark (25 Low, 25 Med, 25 High, 25 Critical), with a False Positive Rate on clean email traffic of only **4.0%** and Critical attack recall of **96.0%**.
- **Mathematical Calibration:** Murphy decomposition confirms low Brier score is driven by high resolution (0.1200) rather than base-rate guessing. Quantile ECE is **0.038 (3.8%)**.

2. End-to-End System Topology & Ingestion Pipeline

The system is organized into six deterministic, decoupled layers executing within an air-gapped environment:



3. Feature Engineering & Feature Noise Cleaning Models

The active feature space (assessment/features.py) is strictly constrained to 8 parsimonious columns (FEATURES_8), enforcing an effective sample size ratio of $p / n_{\text{eff}} = 8 / 272 = 0.029$:

Feature Name	Type / Encoding	Security Function & Invariant Role
version	Native Categorical	Identifies protocol version (TLS 1.0, 1.1, 1.2, 1.3). Catches deprecated RFC 8996 versions.
cipher_strength	Native Categorical	Categorizes cipher suite into weak (RC4, 3DES), medium (CBC), strong (AEAD GCM/ChaCha20).
kex	Native Categorical	Key exchange mechanism (ECDHE, DHE, RSA, None).
chain_valid	Continuous / Boolean	X.509 path validation against CABF roots. Set to 0.5 (neutral) when TLS 1.3 certificate is opaque.
days_to_expiry	Continuous	Normalized remaining certificate lifetime (max(0, days)/365.0). Catches expired/near-expiry certs.
fs_flag	Boolean Flag	Forward Secrecy flag (1.0 = ephemeral Diffie-Hellman, 0.0 = static RSA key exchange).
starttls_mode	Native Categorical	STARTTLS state transition (upgrade, implicit, none, stripped).
miss_indicator_days_to_expiry	Boolean Flag	Missingness indicator ensuring robust performance on passive spans or encrypted TLS 1.3 handshakes.

3.1 Noise Cleaning & Denoising Techniques

- **Zero-Variance Subspace Denoising (`_handle_zero_variance`):** Adds Gaussian perturbation to static feature columns to regularize matrix inversions without distorting discrete signals.
- **Weak Supervision Label Denoising (`FlyingSquid Triplet-Mean`):** Solves pairwise and triplet agreement moments in closed form to denoise heuristic rules without manual labels.
- **Causal Feature Confounder Removal (`CPI & TRIP`):** Conditional Permutation Importance conditions permutations on all other features, eliminating correlation confounder noise at $n \leq 500$.

4. Comprehensive Machine Learning Model Taxonomy

Model Category	Algorithm / Class	Hyperparameters & Config	Primary Operational Role
Supervised Risk	XGBoost Stump	<code>max_depth=1, n_estimators=80, tree_method=hist, enable_categorical=True</code>	Primary active classifier predicting flow vulnerability.
Probability Calibration	Platt Sigmoid (2-Fold CV)	<code>CalibratedClassifierCV(method=sigmoid, cv=2)</code>	Converts raw margins into true posterior probabilities $P(Y=1 x)$.
Label Denoising	FlyingSquid Triplet-Mean	Closed-form moment matching $O(nm)$, cardinality=2	Denoises 23 weak supervision labeling functions.
Supervised Benchmark	CatBoost (CPU)	<code>depth=4-6, l2_leaf_reg=1-3, min_data_in_leaf=1</code>	Symmetric tree baseline for categorical benchmarks.
Foundation Benchmark	TabPFN-v3	Transformer prior-fitted network (N=1000)	External reference benchmark for tabular in-context learning.
Anomaly Tail CDF	PyOD ECOD	Left/Right Empirical CDF tail log-probabilities, <code>contamination=0.10</code>	Non-blocking advisory anomaly tooltip.
Anomaly Copula	PyOD COPOD	Empirical copula multivariate dependence estimator	Models joint multi-dimensional tail correlations.
Anomaly Subspace	Isolation Forest	<code>n_estimators=50, max_samples=min(256, n), contamination=0.10</code>	Tree subspace partitioning robust to discrete step jumps.
Anomaly Ensemble	Z-Score Soft-Voting Hybrid	<code>S_hybrid = 0.20 * z(S_ECOD) + 0.80 * z(S_IForest)</code>	Primary high-precision anomaly detector (89% accuracy).
Active Learning	KMeans-15 + Entropy Selector	k=15 clusters + max-entropy boundary sampling	Queries high-uncertainty flows for human review ($\kappa = 0.81$).
Fingerprint Rarity	JA4 Continuous Density	RFC 8701 GREASE-16 filtering + Internet frequency table in [0, 1]	Quantifies ClientHello fingerprint prevalence.

5. Supervised Risk Assessment: Mathematical Verification & Calibration

Metric Description	Measured Value	Baseline / Threshold	Status
Average Precision (AP)	0.990 [0.974, 0.995]	0.396 (Rule Baseline)	PASS (+0.594)
Leave-One-Family-Out (LOFAM)	0.973	> 0.750	PASS
Grouped Cross-Validation (EnvCV)	0.984	> 0.750	PASS
Generalization Gap	0.010	< 0.050	PASS
Permutation Test p-value	0.0010 (1000 shuffles)	$p < 0.050$	SIGNIFICANT
Brier Score (Prob. Calibration)	0.065 [0.038, 0.089]	0.179 (Base Rate)	NON-OVERLAPPING
Murphy Brier Resolution	0.1200 (Rel: 0.0107, Unc: 0.1786)	RES >> REL	HIGH RESOLUTION
Quantile Expected Calib. Error (ECE)	0.038 (3.8%)	< 0.100	WELL-CALIBRATED

Metric Description	Measured Value	Baseline / Threshold	Status
Smooth Kernel ECE (Nadaraya-Watson)	0.050 (5.0%)	< 0.100	WELL-CALIBRATED

6. Unsupervised Anomaly Detection: Clean Baseline & Multi-Detector Fusion

Traffic Severity Tier	Validation Samples	Anomaly Detection Recall	Per-Class Accuracy
Low Risk (Clean Normal Email)	25 flows	4.0% False Alarms (1/25)	96.0% Correct
Medium Risk (Outdated TLS / CBC)	25 flows	8.0% Anomaly Rate	92.0% Correct
High Risk (Stripping / Expired)	25 flows	72.0% Caught	72.0% Correct
Critical Risk (3DES / RC4 / CVE)	25 flows	96.0% Caught (24/25)	96.0% Correct
★ Multi-Detector Overall Accuracy	100 flows (Balanced)	—	89.0% (Target > 80%)
★ Clean False Positive Rate (FPR)	25 Clean Flows	4.0%	PASS (< 5.0%)

The "Do-Not-Block" Safety Invariant: Anomaly scores are strictly rendered as advisory tooltips for SOC Tier-2 analysts and are mathematically forbidden from automatically dropping or rejecting mail traffic.

7. Comparative Operating Matrix: Rules vs. Risk vs. Anomaly

Operating Quadrant	Supervised Risk	Anomaly Score	Real-World Scenario	Automated Disposition
1. Clean Traffic	Low (P < 0.15)	Low (S < tau)	Gmail / Microsoft 365 standard TLS 1.3 handshake with valid DigiCert root.	allow (Instant Delivery)
2. Known Bad Spec	High (P >= 0.80)	Low (S < tau)	Legacy Postfix server running TLS 1.0 with 3DES Sweet32 cipher suite.	block / quarantine (Drop/Hold)
3. Novel Client / Scraper	Low (P < 0.15)	High (S >= tau)	Automated scraper using RFC-compliant TLS 1.2 but rare elliptic curve & unique JA4.	deliver_banner + SOC Tooltip
4. Active Exploitation	High (P >= 0.80)	High (S >= tau)	STARTTLS stripping downgrade with pre-TLS command injection and custom C2 client.	hold_incident (Immediate Escalation)

8. Adversarial Robustness, Target Shuffling & Generalization Audits

- **Negative Control (Target Shuffling):** Randomly permuting risk labels collapses classifier performance to pure chance (AUC **0.435** [0.293, 0.588]), confirming zero group key leakage.
- **JA4-Ablation Integrity:** Stripping ja4_rarity maintains an AUC of **0.912**, proving the model evaluates multi-dimensional cryptographic features rather than memorizing fingerprint frequency.
- **Design Effect & Sample Audit:** Formal intra-cluster correlation analysis (DEFF = 1.836, ICC = 0.30) establishes an audited effective sample size of n_eff = 272, guaranteeing a strict parsimony ratio of p / n_eff = 0.029.

9. Production Deployment, Verification Suite & Disclosures

Deployment Attribute	Specification & Operational SLA
Container Topology	Single-port (:8000) lightweight Docker appliance serving FastAPI REST endpoints and React 18 SPA.
Model File Sizes	models/risk_clf.pkl (170 KB), models/anomaly.pkl (23 KB). Total storage footprint < 200 KB.
Inference Latency	Supervised risk: < 0.45 ms ; Anomaly scoring: < 0.35 ms ; REST endpoint SLA: < 50 ms .
Test Suite Status	35 / 35 Unit & Integration Tests Passed (100%) across anomaly, calibration, and API wiring suites.
Air-Gap Compliance	100% offline execution. Zero external DNS lookups, zero outbound telemetry, zero private key requirements.

Mandatory Weak Supervision Disclosure: Labels are derived from deterministic weak supervision (`score.py` 23 RFC/CABF checks, 20 scored + 3 info); not hand-labeled field data; `n_eff` = 272 operational distinct. See Dataset Charter §1/§4a.

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